

Then, we can follow the same procedure to define

$$\tilde{\mathbf{d}} = [\mathbf{Q}v]^\top = [v^\top \mathbf{Q}_0 \quad \cdots \quad v^\top \mathbf{Q}_M \quad v^\top \mathbf{Q}_{M+1} \quad \cdots \quad v^\top \mathbf{Q}_{M+L-1}].$$

Again, each row of $\mathbf{D}$ is simply a left-shift version of $\tilde{\mathbf{d}}$. Hence, the main computation cost comes from the matrix-vector multiplication $\tilde{\mathbf{d}} = [\mathbf{Q}v]^\top$, which is not expensive any more.

C Details About RECL

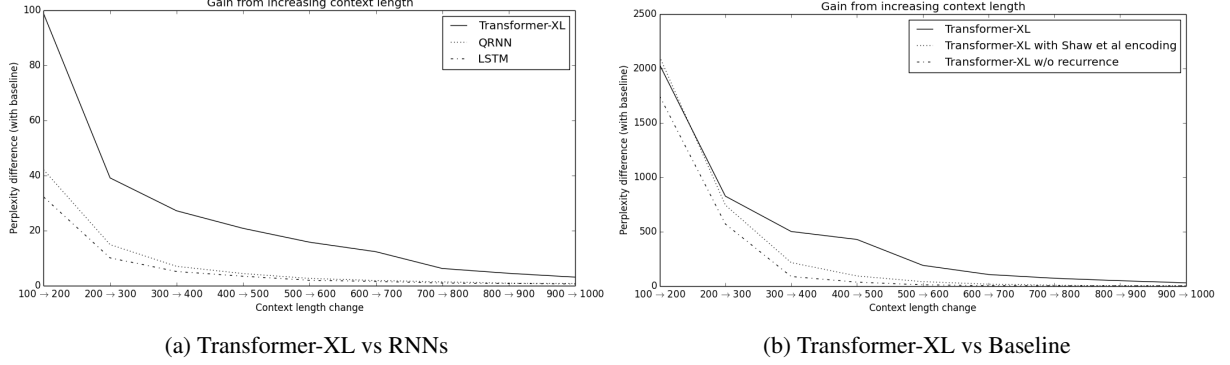


Figure 3: Visualizing unnormalized relative perplexity gains with $r = 0.1$.

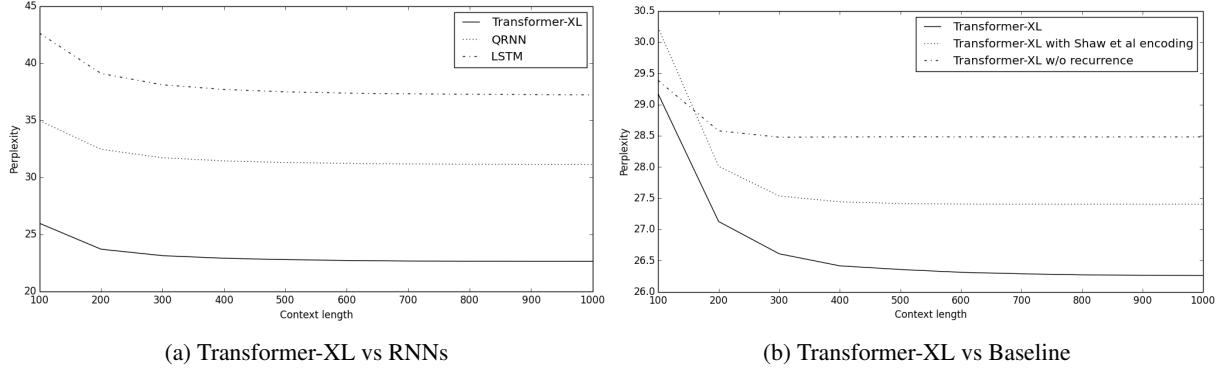


Figure 4: Perplexity vs context length.

In this section, we describe the details of the metric RECL. Let $\mathcal{M} = \{m_1, m_2, \dots, m_N\}$ be a model group consisting of N models. Let $l_i(c, t)$ denote the loss of model m_i on the t -th token in the corpus with a context length c . Concretely, the loss can be written as

$$l_i(c, t) = -\log P_{m_i}(x_t | x_{t-1}, \dots, x_{t-c})$$

where P_{m_i} is the probability distribution given by model m_i , and x_t is the t -th token in the corpus. Given a short context length c and a long context length c' such that $c' \geq c$, we can further define a baseline for each position t ,

$$b(c, t) = \min_{i=1}^N l_i(c, t)$$

The *relative loss* of m_i w.r.t. the model group $\mathcal{M}$ is written as

$$f_i(c, c') = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \min(b(c, t), l_i(c', t))$$

The above equation uses the minimum loss of all models on the short length c as a baseline, and only losses smaller than the baseline will be effectively counted towards the relative loss. This enables fair